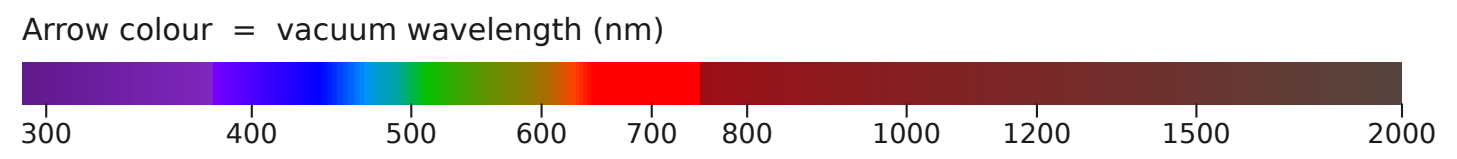
Samarium

Sm (Sm I). Strongest classified transitions below 2 µm (NIST): 67 levels, 90 lines.

Sm$^+$ ionisation limit 45519.69 cm^{-1} = 5.64372 eV (219.685 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|\langle J\|er\|J'\rangle|$ in ea_0
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).


Arrow colour = vacuum wavelength (nm)

even
odd
J = 0
J = 1
J = 2
J = 3
J = 4
J = 5
J = 6
J = 7
J = 8
Energy above the ground state (cm⁻¹) — non-uniform scale
33 000
32 000
31 000
30 500
30 000
29 000
28 000
27 000
26 500
26 000
25 000
24 500
24 000
23 500
23 000
22 500
22 000
21 000
20 000
19 000
18 000
17 000
16 000
15 000
14 000
13 000
12 000
11 000
10 000
9 000
8 000
7 000
6 000
5 000
4 000
3 000
2 000
1 000
0
7F0 0.000 cm⁻¹ stable
7F1 292.554 cm⁻¹
7F2 811.872 cm⁻¹
7F3 1489.5
7F4 2273.5
7F5 3125.41
7F6 4020.62
9H 11877.360 cm⁻¹
9H8 14591.80
(33507)°7 33507.230 cm⁻¹ τ = 20.5 ns
9I°8 31129.740 cm⁻¹ τ = 32.9 ns
5H°7 30330.790 cm⁻¹ τ = 51.8 ns
7D°5 30233.580 cm⁻¹ τ = 25.6 ns
7F°6 30092.414 cm⁻¹ τ = 15.3 ns
(29803)°6 29803.346 cm⁻¹ τ = 34.4 ns
7D°4 29364.983 cm⁻¹ τ = 32 ns
(29070)°5 29069.865 cm⁻¹ τ = 18.5 ns
7D°3 28356.189 cm⁻¹ τ = 29.5 ns
7F°4 28180.913 cm⁻¹ τ = 10.3 ns
7F°3 27425.328 cm⁻¹ τ = 10.1 ns
7G°7 27287.496 cm⁻¹ τ = 8.6 ns
7D°1 26962.184 cm⁻¹ τ = 24.8 ns
(26804)°1 26803.634 cm⁻¹ τ = 67.5 ns
7F°2 26786.751 cm⁻¹ τ = 12.7 ns
(26471)°6 26471.335 cm⁻¹ τ = 16 ns
7F°1 26281.063 cm⁻¹ τ = 11.5 ns
(26181)°6 26180.964 cm⁻¹ τ = 16.5 ns
(25840)°6 25840.296 cm⁻¹ τ = 13.3 ns
(25616)°5 25615.537 cm⁻¹ τ = 11.5 ns
(25391)°5 25390.821 cm⁻¹ τ = 60.1 ns
(25387)°4 25387.153 cm⁻¹ τ = 68.1 ns
(24669)°5 24668.734 cm⁻¹ τ = 10 ns
(24634)°4 24633.707 cm⁻¹ τ = 10 ns
(24467)°3 24467.267 cm⁻¹ τ = 63.4 ns
7D°5 24323.421 cm⁻¹ τ = 12.7 ns
(24312)°4 24311.948 cm⁻¹ τ = 41.3 ns
9H°3 24244.738 cm⁻¹ τ = 38.2 ns
7H°3 24150.732 cm⁻¹ τ = 21.4 ns
(23997)°4 23996.560 cm⁻¹ τ = 22.9 ns
(23734)°7 23734.345 cm⁻¹ τ = 31.5 ns
(23710)°3 23709.922 cm⁻¹ τ = 15.6 ns
7D°4 23594.803 cm⁻¹ τ = 8.7 ns
(23547)°2 23546.469 cm⁻¹ τ = 31.1 ns
7G°2 23380.669 cm⁻¹ τ = 11 ns
7H°7 23337.354 cm⁻¹ τ = 47.5 ns
(23317)°3 23316.546 cm⁻¹ τ = 22.5 ns
9I°3 23230.674 cm⁻¹ τ = 36.2 ns
7F°6 22944.337 cm⁻¹ τ = 27.2 ns
7G°1 22914.043 cm⁻¹ τ = 7.3 ns
(22893)°2 22893.345 cm⁻¹ τ = 27.7 ns
5F°4 22892.940 cm⁻¹ τ = 24.1 ns
9I°2 22843.943 cm⁻¹ τ = 27.5 ns
(22643)°6 22643.100 cm⁻¹ τ = 38 ns
7D°3 22632.269 cm⁻¹ τ = 8.2 ns
5G°2 22491.899 cm⁻¹ τ = 31.4 ns
7F°1 22313.629 cm⁻¹ τ = 11.1 ns
7F°0 22041.017 cm⁻¹ τ = 10.4 ns
7D°2 21813.211 cm⁻¹ τ = 7.7 ns
5F°5 21809.669 cm⁻¹ τ = 33.7 ns
5F°3 21700.876 cm⁻¹ τ = 34.3 ns
7D°1 21193.640 cm⁻¹ τ = 7.5 ns
5F°2 20762.804 cm⁻¹ τ = 43.4 ns
(20713)°4 20712.769 cm⁻¹ τ = 33.3 ns
5F°1 20090.935 cm⁻¹ τ = 47 ns
7G°3 19776.894 cm⁻¹ τ = 38.7 ns
374.958 nm | 1.268 ea₀
380.502 nm | 1.484 ea₀
384.737 nm | 1.051 ea₀
392.631 nm | 2.234 ea₀
377.441 nm | 1.709 ea₀
395.300 nm | 2.642 ea₀
383.389 nm | 1.603 ea₀
372.208 nm | 1.516 ea₀
375.750 nm | 2.183 ea₀
385.567 nm | 1.752 ea₀
397.578 nm | 3.421 ea₀
369.113 nm | 1.711 ea₀
374.652 nm | 2.328 ea₀
381.104 nm | 1.849 ea₀
385.983 nm | 2.717 ea₀
399.114 nm | 3.489 ea₀
381.491 nm | 2.455 ea₀
385.439 nm | 2.424 ea₀
581.649 nm | 4.095 ea₀
383.556 nm | 4.507 ea₀
387.857 nm | 3.004 ea₀
507.810 nm | 5.219 ea₀
528.664 nm | 3.843 ea₀
580.211 nm | 5.002 ea₀
577.537 nm | 6.569 ea₀
422.736 nm | 1.835 ea₀
418.451 nm | 2.025 ea₀
439.459 nm | 1.997 ea₀
441.282 nm | 2.461 ea₀
450.037 nm | 3.272 ea₀
466.486 nm | 3.122 ea₀
438.165 nm | 2.593 ea₀
444.640 nm | 5.044 ea₀
433.735 nm | 4.918 ea₀
432.567 nm | 2.489 ea₀
484.306 nm | 7.734 ea₀
429.796 nm | 8.257 ea₀
458.302 nm | 5.673 ea₀
444.305 nm | 3.525 ea₀
488.534 nm | 6.232 ea₀
439.858 nm | 1.917 ea₀
443.090 nm | 2.938 ea₀
447.214 nm | 4.734 ea₀
428.403 nm | 3.614 ea₀
464.937 nm | 2.647 ea₀
444.352 nm | 2.913 ea₀
436.414 nm | 2.530 ea₀
432.074 nm | 3.150 ea₀
433.123 nm | 2.885 ea₀
428.470 nm | 2.383 ea₀
442.058 nm | 2.883 ea₀
452.444 nm | 1.644 ea₀
459.803 nm | 2.146 ea₀
448.157 nm | 1.939 ea₀
465.078 nm | 3.014 ea₀
467.206 nm | 2.503 ea₀
468.287 nm | 2.435 ea₀
453.507 nm | 2.899 ea₀
450.464 nm | 2.119 ea₀
428.340 nm | 4.513 ea₀
446.054 nm | 2.276 ea₀
475.206 nm | 2.294 ea₀
471.743 nm | 5.214 ea₀
451.257 nm | 2.893 ea₀
458.287 nm | 2.399 ea₀
467.215 nm | 2.887 ea₀
484.968 nm | 3.354 ea₀
464.670 nm | 2.647 ea₀
476.160 nm | 4.064 ea₀
492.036 nm | 3.403 ea₀
471.840 nm | 2.482 ea₀
478.444 nm | 3.296 ea₀
490.635 nm | 2.173 ea₀
472.975 nm | 4.339 ea₀
491.178 nm | 4.663 ea₀
488.514 nm | 2.481 ea₀
478.721 nm | 2.794 ea₀
469.004 nm | 3.883 ea₀
440.240 nm | 3.556 ea₀
497.737 nm | 1.818 ea₀
492.543 nm | 3.067 ea₀
445.420 nm | 3.532 ea₀
527.287 nm | 3.044 ea₀
520.203 nm | 4.207 ea₀
511.859 nm | 4.483 ea₀
512.356 nm | 3.941 ea₀
504.568 nm | 4.111 ea₀
528.437 nm | 3.928 ea₀
517.686 nm | 4.655 ea₀
507.261 nm | 5.535 ea₀

Strongest transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$\|\langle J\|er\|J'\rangle\|$ (ea₀)	A (s⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
$^7F_6$ – $^7F^o_6$	383.5563	383.4475	781.6127	4.507	5.610e+07	10.4	85.8	
$^7F_2$ – $^7F^o_1$	392.6312	392.5201	763.5471	2.234	5.570e+07	13.84	64.1	
$^7F_4$ – $^7F^o_3$	397.5781	397.4657	754.0467	3.421	5.390e+07	15.76	54.4	
$^7F_6$ – $^7G^o_7$	429.7956	429.6747	697.5234	8.257	1.160e+08	18.51	99.8	
$^7F_0$ – $^7G^o_1$	436.4136	436.2910	686.9457	2.530	5.200e+07	21.8	38	
$^7F_1$ – $^7G^o_1$	442.0575	441.9334	678.1752	2.883	6.500e+07	21.8	47.4	
$^7F_5$ – $(25616)^o_5$	444.6397	444.5149	674.2369	5.044	5.330e+07	13.84	61.3	
$^7F_4$ – $(24634)^o_4$	447.2138	447.0883	670.3560	4.734	5.640e+07	15.92	56.4	
$^7F_6$ – $(25840)^o_6$	458.3020	458.1736	654.1374	5.673	5.210e+07	11.97	69.3	
$^7F_1$ – $^7F^o_0$	459.8026	459.6738	652.0025	2.146	9.600e+07	15.3	99.8	
$^7F_2$ – $^7F^o_1$	465.0783	464.9481	644.6065	3.014	6.100e+07	14.34	67.7	
$^7F_2$ – $^7D^o_2$	476.1601	476.0270	629.6043	4.064	6.200e+07	20.67	47.7	
$^7F_1$ – $^7D^o_1$	478.4440	478.3103	626.5988	3.296	6.700e+07	21.22	50.2	
$^7F_6$ – $(24669)^o_5$	484.3058	484.1705	619.0149	7.734	9.700e+07	15.92	97	
$^7F_5$ – $^7D^o_4$	488.5344	488.3980	613.6568	6.232	7.500e+07	18.29	65.2	
$^7F_4$ – $^7D^o_3$	491.1777	491.0406	610.3544	4.663	5.310e+07	19.41	43.5	

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

^{147}Sm hyperfine structure (I = 7/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
$^7F_1$	-33.493	-58.688	F=5/2: +119.28 F=7/2: +75.41 F=9/2: -131.90
$^7G^o_1$	-212.620	-9.630	F=5/2: +951.63 F=7/2: +219.50 F=9/2: -746.58

Measured A, B constants; sources in the data file.